Assignment No.: 7

Title:

Practical Based on SQL Subqueries (Using AdventureWorks2019)

Objective:

To study and implement **SQL subqueries** using the AdventureWorks database.

Database Used:

`AdventureWorks2019`

Key Tables Used:

- `HumanResources.Employee`
- `Person.Person`
- `HumanResources.Department`
- `Sales.SalesOrderHeader`

Programs with Output

1) Subquery using `IN`

Query:

```
USE AdventureWorks2019;

SELECT BusinessEntityID, JobTitle
FROM HumanResources.Employee
WHERE BusinessEntityID IN (
    SELECT BusinessEntityID
    FROM Sales.SalesPerson
);
```

Output (Sample):

```
BusinessEntityID   JobTitle
----------------   ------------------------
274                Sales Representative
275                Sales Representative
...
```

2) Subquery using = (Single-row subquery)

Query:

```
SELECT BusinessEntityID, JobTitle
FROM HumanResources.Employee
WHERE BusinessEntityID = (
    SELECT TOP 1 BusinessEntityID
    FROM HumanResources.Employee
    ORDER BY HireDate
);
```

Output (Sample):

```
BusinessEntityID   JobTitle
----------------   -------------------------
28                  Production Technician - WC60
```

3) Subquery using ANY

Query:

```
SELECT BusinessEntityID, JobTitle
FROM HumanResources.Employee
WHERE BusinessEntityID > ANY (
    SELECT BusinessEntityID
    FROM HumanResources.Employee
    WHERE JobTitle LIKE 'Sales%'
);
```

Output (Sample):

```
BusinessEntityID   JobTitle
----------------   -------------------------
280                Engineer
281                Manager
...
```

4) Subquery using ALL

Query:

```
SELECT BusinessEntityID, JobTitle
FROM HumanResources.Employee
WHERE BusinessEntityID > ALL (
    SELECT BusinessEntityID
    FROM HumanResources.Employee
    WHERE JobTitle LIKE 'Sales%'
);
```

Output (Sample):

```
BusinessEntityID   JobTitle
----------------   -------------------------
290                Design Engineer
```

5) Correlated Subquery

Query:

```sql
USE AdventureWorks2019;

SELECT e.BusinessEntityID, e.JobTitle
FROM HumanResources.Employee e
WHERE e.HireDate > (
    SELECT MIN(e2.HireDate)
    FROM HumanResources.Employee e2
);
```

Output (Sample):

```
BusinessEntityID   JobTitle
----------------   -------------------------
260                Tool Designer
261                Production Technician
262                Production Supervisor
...
```

Conclusion:

This practical demonstrates the use of SQL subqueries with operators such as `IN`, `ANY`, `ALL`, and correlated subqueries using the AdventureWorks2019 database.